Computational Analysis of the ZWINT Gene Mutational Landscape in Human Cancers

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Biological Datasets for Computational Physics

The characterization of somatic mutations in cancer-related genes is fundamental for understanding tumor heterogeneity and systemic genomic instability. This study presents a computational workflow evaluating the mutational landscape of the *ZWINT* (ZW10 interacting kinetochore protein) gene using data exclusively retrieved from the Catalogue of Somatic Mutations in Cancer (COSMIC) database. Data retrieval, filtering, and statistical analysis was performed during this project. Our findings show that *ZWINT* mutations are predominantly concentrated in high-turnover carcinomas, specifically the lung ($n = 60$), large intestine ($n = 50$), and stomach ($n = 41$). Sequence feature analysis indicates a high prevalence of substitution-missense mutations ($n = 176$). Furthermore, spatial sequence mapping reveals an overall diffuse distribution of variants across the protein sequence, with one statistically robust exception: residue 187, which accumulates mutations far in excess of chance expectation ($p < 10^{-4}$). These results suggest that ZWINT's mutational landscape combines a background of passenger mutations driven by systemic chromosomal instability with at least one candidate mutational hotspot warranting further investigation.

I. INTRODUCTION

A. Biological problem: what is known and what remains unknown

The kinetochore is the macromolecular structure that couples chromosomes to spindle microtubules during mitosis, and its outer layer mediates microtubule attachment and force transduction during chromosome segregation [1]. ZWINT (ZW10 interacting kinetochore protein) is a metazoan-specific component that is required for correct kinetochore–microtubule attachment. Loss of ZWINT function has been shown to disrupt kinetochore-microtubule attachment and chromosome alignment, leading to aneuploidy. [2].

Independently of its structural role, ZWINT mRNA and protein are recurrently overexpressed across a broad panel of solid tumors, including lung, breast, colorectal, hepatocellular and pancreatic carcinomas [3–8], and this overexpression is consistently associated with worse overall survival, higher recurrence and increased metastatic risk [3–6]. What remains comparatively unexplored is whether this oncogenic association is accompanied by a distinctive *mutational* signature at the DNA level, as opposed to a purely transcriptional dysregulation. In other words: is ZWINT mutated in a way that suggests positive selection for specific functional variants (a driver-like pattern), or is its mutational burden better explained as a byproduct of the same chromosomal instability (CIN) that its own dysfunction is known to promote [9], given that the large majority of somatic mutations detected in tumor genomes are known to be passenger rather than driver events?[10, 11]

B. Defining the question

This project addresses the following question: *does the somatic mutational landscape of ZWINT, as catalogued in COSMIC, show evidence of tissue-specific concentration, a dominant functional mutation class, or spatial clustering along the protein sequence that would be consistent with positive selection – or is it instead compatible with a diffuse, passenger-like pattern expected under systemic genomic instability?*

C. Defining the plan

To answer this, the biological question was translated into three computational sub-tasks operating on tabular COSMIC exports[12, 13]: (i) grouping and counting mutation records by primary tissue to test for anatomical concentration; (ii) categorizing records by annotated sequence-consequence type to quantify the functional-class composition of the variant pool; and (iii) mapping variant counts onto integer positions along the ZWINT protein sequence to test for spatial clustering versus uniform dispersion. Each subtask is paired with a corresponding statistical test (Section 2.3) so that qualitative visual patterns can be supported quantitatively rather than assessed by inspection alone.

II. METHODS

A. Dataset Assembly and Filtering

In order to maintain consistency and work on all the tasks in the same framework, all data were retrieved exclusively from the Catalogue of Somatic Mutations in Cancer (COSMIC) database [12], in-

cluding annotations from the Cancer Gene Census [14]. Two primary datasets were processed:

- `cosmic-gm.csv`: Utilized for general metadata, sample histology classifications, and primary tissue origins.
- `cosmic-variants.csv`: Utilized for targeted variant statistics, sequence changes (CDS and amino acid alterations), mutation types, and protein sequence positions.

B. Computational Workflow

Data parsing, cleaning, and statistical aggregation were implemented in Python utilizing the `pandas` library. The data filtering pipeline isolated entries specific to the *ZWINT* gene. Categorical variables (such as primary tissue and mutation type) were grouped and aggregated to calculate absolute frequencies. Visualizations were generated using `matplotlib` to map tissue distributions, categorical feature counts, and 1D positional scatter plots along the protein sequence.

C. Statistical analysis

To complement the descriptive frequency counts, two simple chi-squared goodness-of-fit tests were applied against the null hypothesis of a uniform distribution: one on the tissue-level counts underlying Figure 1, and one on the mutation-type counts underlying Figure 2. Both tests use $\alpha = 0.05$ and were computed directly on the aggregated frequency tables produced in Section 2.2, without additional modelling assumptions.

To assess whether mutations cluster along the ZWINT protein sequence (Figure 4), a Monte Carlo permutation test was used: the total number of observed mutations was redistributed uniformly at random across the protein sequence length (10^4 resamples, multinomial model), and the maximum mutation count reached at any single position in each resample was recorded to build a null distribution. The observed maximum count at a single position was then compared against this null distribution to obtain an empirical p-value. A significantly higher observed maximum than expected under the null indicates a non-random local accumulation (a hotspot); a comparable maximum supports a diffuse, passenger-like distribution. This test was preferred over the chi-squared approach here because mutation position is a continuous coordinate rather than a small set of categories, making a resampling-based test more natural and easier to justify than an arbitrary binning scheme.

III. RESULTS

A. Tissue Distribution

The anatomical distribution of *ZWINT* somatic mutations was evaluated by filtering for carcinoma histologies. The resulting frequency distribution demonstrates a clear concentration of mutations in organs characterized by high cellular turnover and epithelial proliferation. As illustrated in Figure 1, the highest mutation frequencies were observed in the **Lung** ($n = 60$), followed by the **Large Intestine** ($n = 50$) and the **Stomach** ($n = 41$). Other anatomical sites exhibited substantially lower mutational counts. This concentration is consistent with the elevated replication-associated mutability of high-turnover epithelial tissues (χ^2-test, $p < 0.001$; Table I). Taken together, these results indicate that the anatomical distribution of ZWINT mutations tracks known high-mutability carcinoma types rather than a single tissue-specific driver role.

B. Sequence Features

A second question is whether the different functional consequence types annotated for ZWINT variants occur with comparable frequency, or whether one category dominates the mutational spectrum (Section 1.2). Mutations were categorized by their annotated sequence-consequence type using the variant table described in Section 2.1, and the resulting frequencies were tested against a uniform-distribution null using a chi-squared test (Section 2.3).

The distribution is markedly non-uniform (Table I): as shown in Figure 2, Substitution-Missense mutations are the dominant annotated category, with 176 occurrences, more than the next two categories combined. A substantial fraction of records (n = 124) remain classified as “Unknown” due to current database annotation constraints, while Substitution-Coding Silent mutations account for 47 cases, and Substitution-Nonsense and Nonstop-extension events are comparatively rare. This basically tells us that most ZWINT variants keep the same reading frame: missense changes alter a single residue without truncating the protein, unlike nonsense events, which would be expected to break protein function completely. In summary, these results show that the ZWINT mutational spectrum is more dominated by missense substitutions than by loss-of-function events, which is consistent with a mutational profile where the coding sequence stays mostly intact – also compatible, though not fully proving it, with a predominance of passenger over driver-like alterations. (Section 4).

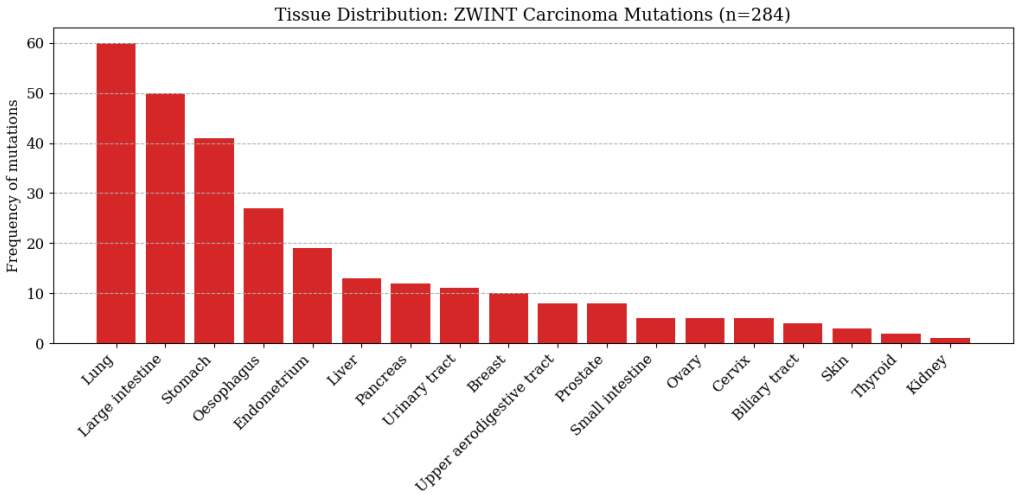


FIG. 1: *Frequency distribution of ZWINT somatic mutations grouped by Primary Tissue.*

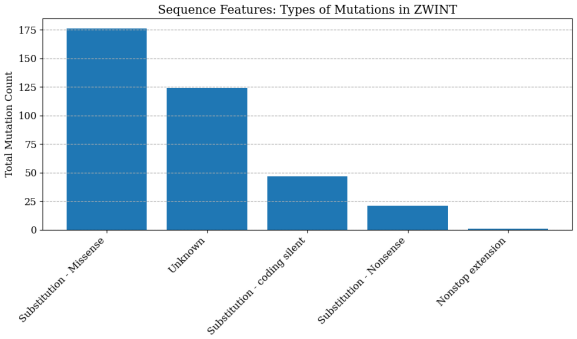


FIG. 2: *Categorical breakdown of ZWINT mutation types from the COSMIC dataset.*

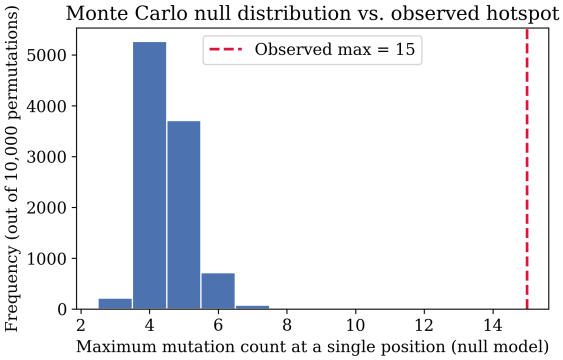


FIG. 3: *Monte Carlo null distribution of the maximum mutation count at a single position under a uniform random model (10^4 resamples)*

C. Distribution and Sequence Alteration

Beyond the overall shape of the positional profile, a specific question is whether any individual residue accumulates more mutations than expected under a uniform random model along the ZWINT protein sequence (Section 1.2). Mutation counts were mapped onto integer positions using the variant table described in Section 2.1, and the observed maximum count at a single position was compared to the null distribution obtained by randomly redistributing the same total number of mutations across the sequence length (10^4 Monte Carlo resamples; Section 2.3).

The positional profile (Figure 4) is overall diffuse, with most positions showing between 1 and 4 mutations, consistent with a background of uniformly scattered variants. However, one position (residue 187) carries 15 mutations, a count that is never reached in 10^4 random redistributions of the same total mutation load ($p < 10^{-4}$; null mean maximum = 4.52).This indicates that, while the bulk of ZWINT variants are compatible with a passenger-like, dispersion-driven pattern, residue 187 constitutes a statistically robust local exception rather than a product of chance clustering. Taken together, these results show that ZWINT's mutational landscape is not uniformly diffuse as a simple visual inspection might suggest, but instead combines a broad passenger-like background with at least one focal, non-random accumulation point that merits closer biological interpretation.

TABLE I: *Chi-squared goodness-of-fit tests against a uniform-distribution null hypothesis, for tissue and mutation-type frequency distributions*

Test	χ^2	d.f.	p-value
Tissue distribution	326.86	17	3.14×10^{-59}
Mutation type	295.00	4	1.30×10^{-62}

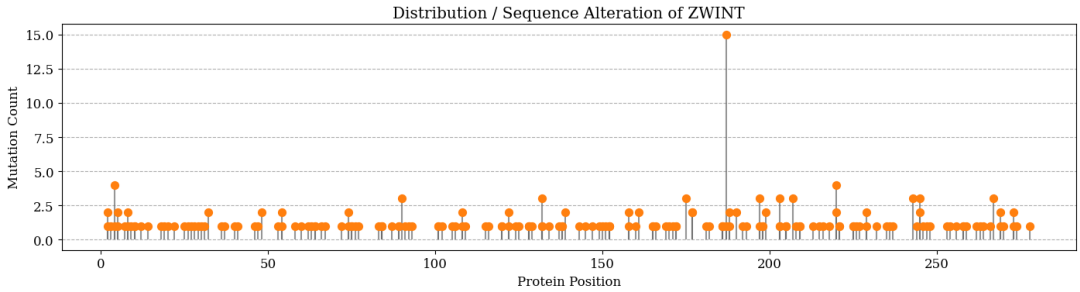


FIG. 4: *Positional mapping and frequency scatter of ZWINT mutations across the protein sequence. The distribution is overall diffuse, with one statistically significant local exception at residue 187 (see Figure 3).*

IV. DISCUSSION AND CONCLUSION

This project asked whether the somatic mutational landscape of ZWINT, as catalogued in COSMIC, shows evidence of positive selection or is instead compatible with a diffuse, passenger-like pattern (Section 1.2). The three analyses converge on a complex answer: ZWINT mutations are strongly concentrated in high-turnover carcinomas – lung, large intestine and stomach – significantly deviating from a uniform tissue distribution (χ^2-test, $p < 0.001$; Table I; Section 3.1); substitution-missense dominates the functional spectrum (Section 3.2); and while most variants are diffusely distributed along the protein sequence, one residue (p.R187) is a statistically robust exception, recurring far in excess of chance expectation ($p < 10^{-4}$; Section 3.3).

The tissue-level concentration agrees with the chromosomal instability (CIN) framework from Carter et al. [9], where genes involved in chromosome segregation, such as ZWINT, drive genomic instability instead of acting individually as classical drivers. The p.R187G recurrence, on the other hand, meets the standard criteria used in cancer genomics to flag a hotspot; however, Hess et al. [15] also pointed out that some hotspots can appear simply because of a locally higher background mutability, rather than actual selection. Without doing any further analysis of that local mutability, residue 187 is better described as a candidate hotspot rather than confirmed as a driver, which leads to a two-part model for the ZWINT landscape: a passenger background with one statistically robust, non-random substitution on top of it.

Resolving the functional status of p.R187G would require mapping it onto 3D structural models of ZWINT and a sequence-context-aware mutation model [15]. In general, it would be interesting for future work to relate this mutational profile to ZWINT's already known overexpression across several tumor types, to see whether the two go together.

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